

NHANES Data Analysis Project

Python Data Analysis and Visualization Documentation

Prepared by: Dimple A

1. Objective

The objective of this project is to analyze NHANES adult body measurement data using Python. The project loads male and female measurement datasets, cleans missing values, visualizes weight distribution, calculates BMI, compares waist-to-height ratios, and displays the top five standardized female records based on BMI.

2. Technologies Used

- Python 3
- Pandas for reading and handling CSV data
- NumPy for numerical calculations and array operations
- Matplotlib for data visualization
- SciPy for statistical analysis support
- Google Colab / Jupyter Notebook for running the program

3. Dataset Details

The program uses NHANES adult body measurement CSV files. Separate datasets are used for male and female records. The data is loaded directly from online CSV links using pandas `read_csv()`. Missing values are removed using `dropna()` before analysis.

Dataset	Purpose	Source Type
Male BMI measurements	Used for male weight and waist-to-height ratio comparison	Online CSV
Female BMI measurements	Used for female weight, BMI calculation, standardization, and ratio comparison	Online CSV

4. Project Workflow

1. Import required Python libraries.
2. Connect to NHANES male and female CSV data links.
3. Load the datasets using pandas and remove missing values.
4. Convert pandas DataFrames into NumPy arrays for numerical processing.
5. Create histograms to compare female and male weight distributions.
6. Calculate BMI for female records using the formula: $\text{weight} / \text{height squared}$.
7. Standardize female data using z-score standardization.
8. Calculate waist-to-height ratio for both male and female records.
9. Create a boxplot to compare male and female waist-to-height ratios.

10. Sort female records based on BMI and print the top five standardized values.

5. Important Formulae

BMI Formula: $BMI = \text{weight in kilograms} / (\text{height in meters})^2$

Waist-to-Height Ratio Formula: $\text{waist circumference} / \text{height}$

Z-score Standardization Formula: $z = (\text{value} - \text{mean}) / \text{standard deviation}$

6. Source Code

```
import pandas as pd
import numpy as np
import matplotlib.pyplot as plt
from scipy import stats

print("Step 1: Connecting to NHANES data...")
male_url = "https://raw.githubusercontent.com/gagolews/teaching-data/master/marek/nhanes_adult_male_bmx_2020.csv"
female_url = "https://raw.githubusercontent.com/gagolews/teaching-data/master/marek/nhanes_adult_female_bmx_2020.csv"

# Load data and clean it
male_df = pd.read_csv(male_url).dropna()
female_df = pd.read_csv(female_url).dropna()
male = male_df.values
female = female_df.values

print("Step 2: Generating Weight Comparison...")
plt.figure(figsize=(10, 5))
plt.subplot(1, 2, 1)
plt.hist(female[:, 0], bins=20, color='pink', edgecolor='black')
plt.title('Female Weights')
plt.subplot(1, 2, 2)
plt.hist(male[:, 0], bins=20, color='skyblue', edgecolor='black')
plt.title('Male Weights')
plt.show()

print("Step 3: Calculating BMI...")
f_bmi = female[:, 0] / ((female[:, 1] / 100) ** 2)
female = np.column_stack((female, f_bmi))
zfemale = (female - np.mean(female, axis=0)) / np.std(female, axis=0)

print("Step 4: Body Ratios (Waist-to-Height)...")
f_ratio = female[:, 6] / female[:, 1]
m_ratio = male[:, 6] / male[:, 1]
plt.boxplot([f_ratio, m_ratio], labels=['Female Ratio', 'Male Ratio'])
plt.title('Waist-to-Height Comparison')
plt.show()

print("Step 5: Final Standardized Data (Top 5 BMI):")
sorted_idx = np.argsort(female[:, 7])
print(pd.DataFrame(zfemale[sorted_idx[-5:]], columns=list(male_df.columns)+['BMI']))

print("\n--- DONE! You can now share your link! ---")
```

7. Expected Output

- A histogram showing female weight distribution.
- A histogram showing male weight distribution.
- A boxplot comparing female and male waist-to-height ratios.
- A printed table showing the top five standardized female records based on BMI.
- Completion message printed at the end of execution.

8. Output Screenshots

Add your Google Colab / Jupyter Notebook screenshots below this section before final submission if required by your internship portal. Recommended screenshots: code cell, weight histogram output, waist-to-height boxplot output, and final top five BMI table output.

9. Learning Outcome

Through this project, I learned how to load real-world CSV data, clean missing values, perform numerical calculations using NumPy, calculate BMI and body measurement ratios, standardize data, and visualize comparisons using Matplotlib. This project also improved my understanding of basic data analysis workflow in Python.

10. Conclusion

This project successfully demonstrates data loading, preprocessing, visualization, BMI calculation, ratio comparison, and standardized output generation using Python. It shows how Python libraries can be used to analyze real-world health-related datasets in a simple and structured way.

NHANES Data Analysis Project - Output Section

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11. Actual Output

The following output is produced after running the Python code in Google Colab / Jupyter Notebook. The code loads NHANES male and female body measurement datasets, creates visualizations, calculates BMI, compares body ratios, and prints the final standardized BMI result table.

```
Step 1: Connecting to NHANES data...
Step 2: Generating Weight Comparison...

[Output Graph 1]
A figure is displayed with two histograms:
- Left histogram: Female Weights
- Right histogram: Male Weights

Step 3: Calculating BMI...
Step 4: Body Ratios (Waist-to-Height)...

[Output Graph 2]
A boxplot is displayed comparing:
- Female Ratio
- Male Ratio

Step 5: Final Standardized Data (Top 5 BMI):

[Output Table]
A pandas DataFrame table is printed showing the top 5 female records
based on BMI after z-score standardization.

The table contains standardized values for the original body
measurement columns and the calculated BMI column.

--- DONE! You can now share your link! ---
```

12. Output Explanation

Output Part	Meaning
Weight histograms	Shows female and male weight distributions separately.
BMI calculation	Calculates BMI using weight / height squared.
Waist-to-height boxplot	Compares female and male waist-to-height ratios.
Top 5 BMI table	Shows the top five female BMI records after z-score standardization.
Done message	Shows that the program executed successfully.

13. Final Result

The project runs successfully and demonstrates data loading, data cleaning, BMI calculation, ratio comparison, visualization using Matplotlib, and standardized result generation using NumPy and pandas.

Note: The exact numerical table values may differ slightly depending on the dataset version loaded from the online CSV source at runtime.